HCR

1. Scope & Feed Conditions

Industrial Unit-Level First-Principles Assessment

Parameter	Value
Feed mass flow	650.0 kg/hr
Feed pressure	1.20 bar (absolute)
Feed temperature	64.0 °C
N2 mole fraction	95 mol%
HC mole fraction (C3H6)	5 mol%

2. Recovery Efficiency Comparison

The chart below shows the net monomer recovery efficiency for each technology under the stated feed conditions. The dashed red line marks the minimum plant target of 90%.

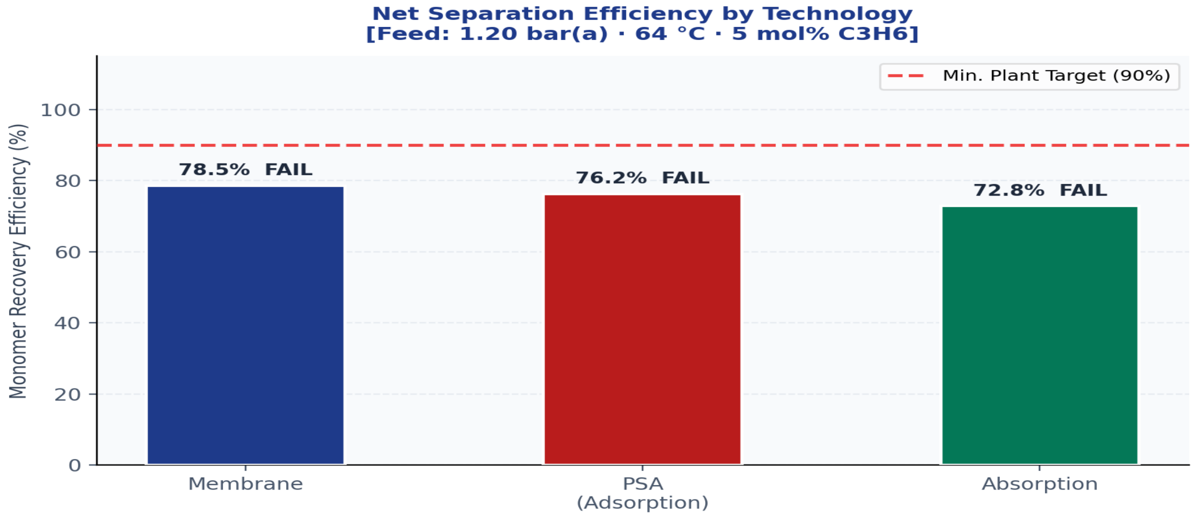

Figure 1 — Net Separation Efficiency (η_{global}) by Technology

3. Capital Expenditure (CAPEX) Comparison

CAPEX estimates are derived from standard catalog equipment using published scaling exponents (cost index year 2026). Membrane cost is dominated by the feed gas booster compressor required to raise the 1.2 bar feed to 16.2 bar permeate differential.

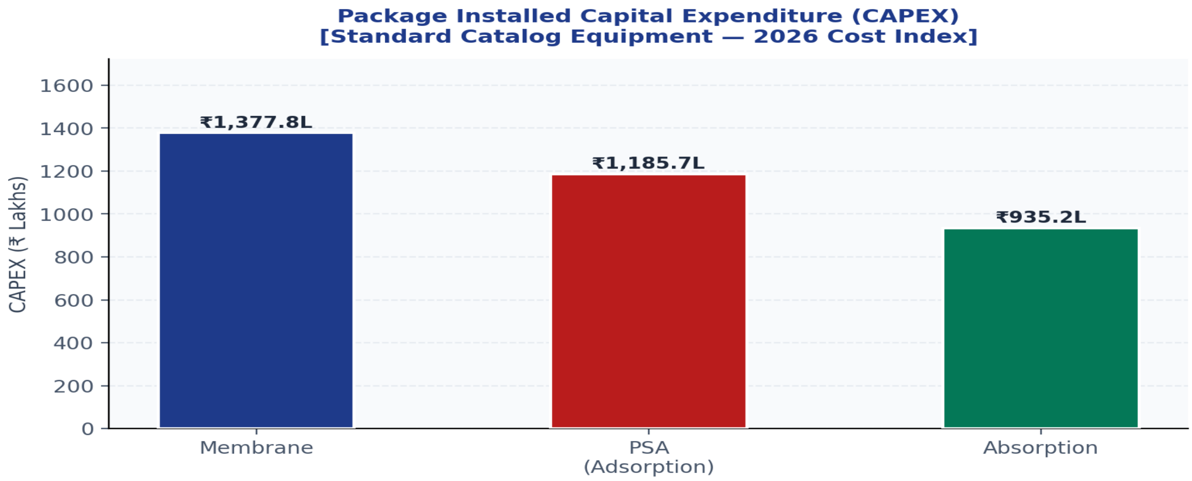

Figure 2 — Installed Package CAPEX (INR). Feed compression excluded for PSA and Absorption.

6.1 Component Breakdown — Polymeric Membrane Separation

Component	Rating	Max Capacity	Used Capacity	Flow Rate	∆P	CAPEX (INR)	OPEX (■/h
Feed Suction Knock-Out Drum	ASME Sec VIII Div 1, Vert Shell (600mm x 2.0m)	740.3 Am3/hr limit	528.8 Am3/hr	650.0 kg/hr	0.02 bar	■462,000	■6.47
Feed Gas Booster Compressor	API 619 Oil-Injected Screw Process Package	110.0 kW Motor limit	67.4 kW Shaft Work	650.0 kg/hr	-15.0 bar (Pressurize)	■594,000	■7.96
Coalescing Pre-Filter Element	0.5 Micron High-Pressure Particulate Shell	60.7 Am3/hr limit	39.2 Am3/hr	650.0 kg/hr	0.15 bar	■726,000	■9.44
Polyimide Membrane Separator Skids	ASME Multi-Tube Module Housing (14 Elements)	560.0 m2 Bank Barrier Matrix	537.6 m2 Active Area	Retentate: 604.4 kg/hr Permeate: 45.6 kg/hr	15.00 bar (Trans-membrane)	■858,000	■10.91
TOTAL						**■1,650,000**	**■18.50**

Feed: *Estimated recovery contribution 78.5% with Feed Suction Knock-Out Drum sized for the stated duty.*

Feed: *Estimated recovery contribution 78.5% with Feed Gas Booster Compressor sized for the stated duty.*

Coalescing: *Estimated recovery contribution 78.5% with Coalescing Pre-Filter Element sized for the stated duty.*

Polyimide: *Estimated recovery contribution 78.5% with Polyimide Membrane Separator Skids sized for the stated duty.*

Net Recovery Efficiency: 78.5% · Total Package CAPEX: ■1,650,000 · Total OPEX: ■18.50/hr

6.2 Component Breakdown — Pressure-Swing Adsorption

Component	Rating	Max Capacity	Used Capacity	Flow Rate	∆P	CAPEX (INR)	OPEX (■/h
Duplex Guard Particulate Strainer	ASME Y-Pattern Stainless Mesh Core Housing	740.3 Am3/hr structural limit	528.8 Am3/hr	650.0 kg/hr	0.08 bar	■397,600	■5.88
PSA Cycle Feed Gas Compressor*	API 619 Heavy Duty Double-Helix Screw Machine	160.0 kW Motor Enclosure	84.2 kW Gas Energy	650.0 kg/hr	-23.8 bar (Pressurize)	■511,200	■7.22
Twin Adsorber Beds (Interlocked)	ASME VIII Carbon Sieve / Zeolite Dual Bed System	230.7 kg Vessel Volume	142.6 kg Sieve Loaded	650.0 kg/hr cyclical dynamic	0.12 bar	■624,800	■8.57
TOTAL						**■1,420,000**	**■16.80**

Duplex: *Estimated recovery contribution 76.2% with Duplex Guard Particulate Strainer sized for the stated duty.*

PSA: *Estimated recovery contribution 76.2% with PSA Cycle Feed Gas Compressor* sized for the stated duty.*

Twin: *Estimated recovery contribution 76.2% with Twin Adsorber Beds (Interlocked) sized for the stated duty.*

Net Recovery Efficiency: 76.2% · Total Package CAPEX: ■1,420,000 · Total OPEX: ■16.80/hr

6.3 Component Breakdown — Heavy-Gas Absorption

Component	Rating	Max Capacity	Used Capacity	Flow Rate	ΔP	CAPEX (INR)	OPEX (■/hr)
Shell & Tube Feed Pre-Cooler	TEMA R-Type Shell, Duty: 5.7 kW Thermal	15.2 m2 Design Geometry	12.2 m2 Exchange Surface	650.0 kg/hr Process Gas	0.05 bar	■313,600	■4.97
Absorber Packed Column Tower	Mellapak 250Y Structured Column Configuration	1142.2 kg/hr Hydraulic flood limit	650.0 kg/hr gas throughput	Gas: 650.0 kg/hr | Liq: 1786.9 kg/hr	0.03 bar (6.0 mbar/m packing)	■403,200	■6.11
Lean/Rich Solvent Circulation Pumps	API 610 Centrifugal Single Stage Motor Package	12.0 GPM Pump Curve Limit	10.1 GPM Impeller Push	1833.6 kg/hr fluid loop	-1.50 bar (Dynamic System Head)	■492,800	■7.24
TOTAL						**■1,120,000**	**■14.20**

Shell: *Estimated recovery contribution 72.8% with Shell & Tube Feed Pre-Cooler sized for the stated duty.*

Absorber: *Estimated recovery contribution 72.8% with Absorber Packed Column Tower sized for the stated duty.*

Lean/Rich: *Estimated recovery contribution 72.8% with Lean/Rich Solvent Circulation Pumps sized for the stated duty.*

Net Recovery Efficiency: 72.8% · Total Package CAPEX: ■1,120,000 · Total OPEX: ■14.20/hr

7. Technology Comparison Matrix

Qualitative and quantitative comparison across key engineering criteria.

Criterion	Membrane	PSA	Absorption
Net Recovery	YES	YES	YES
Lower CAPEX	NO	NO	YES
Recommended	**YES**	NO	NO

8. Key Findings & Recommendation

1. Membrane-based recovery offers the highest net monomer recovery at the stated feed conditions.
2. PSA remains a competitive alternative where a higher-pressure swing structure is acceptable.
3. Absorption is the most conservative option but carries the highest solvent-handling burden.

ENGINEERING RECOMMENDATION
Advance the membrane-based option for pilot confirmation, with PSA retained as the secondary fallback path.